

Final Aerodynamics Class Project: Section II

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Nomenclature

CL:	Coefficient of Lift
CD:	Coefficient of Drag
CDi:	Induced Drag Coefficient
CM:	Moment Coefficient
Cm _{c/4} :	Moment Coefficient at the quarter chord
CP :	Coefficient of Pressure
AoA:	Angle of Attack
AoA _{L=0} :	Angle of Attack at 0 Lift
Re:	Reynolds Number
AR:	Aspect Ratio
S:	Wing Area
b:	Wingspan
c:	Chord length
cr:	Root chord length
ct:	Tip chord length

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A Category 4 hurricane has devastated a remote Caribbean Island, destroying infrastructure and leaving survivors stranded. Your team must design an aircraft to deliver emergency medical supplies and evacuate critically injured individuals to Raleigh, NC. The aircraft's original wing was damaged in the storm, requiring a rapid redesign. Using aerodynamic principles and computational tools, you must optimize a new wing for three flight phases: climb (to avoid storm remnants), cruise (efficiency for long-range rescue), and descent (safe landing in Raleigh).

Part 1

In section one of this project, we went over our mission parameters and found three separate airfoils that could be used for this mission. Below is the comparison of each airfoil during the different stages of flight: climb, cruise, and descent.

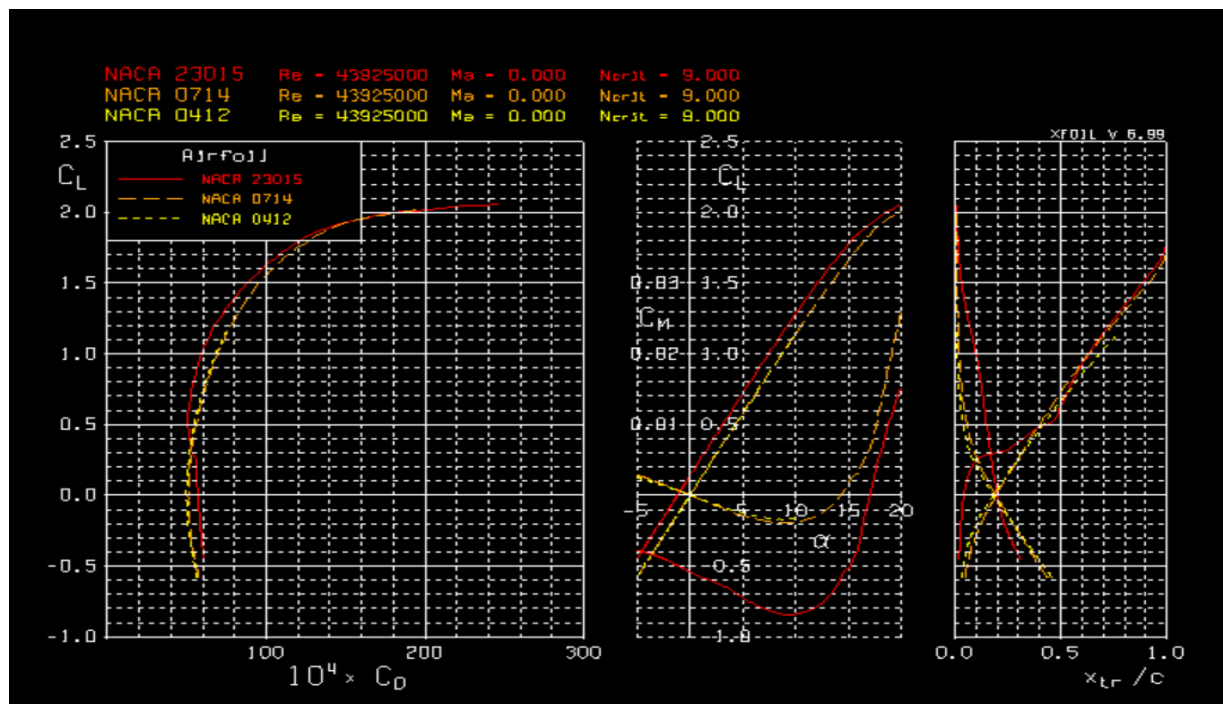


Figure 1: C_L vs. C_D , C_L vs. α for all three airfoils at $Re = 4.4e7$ (Climb)

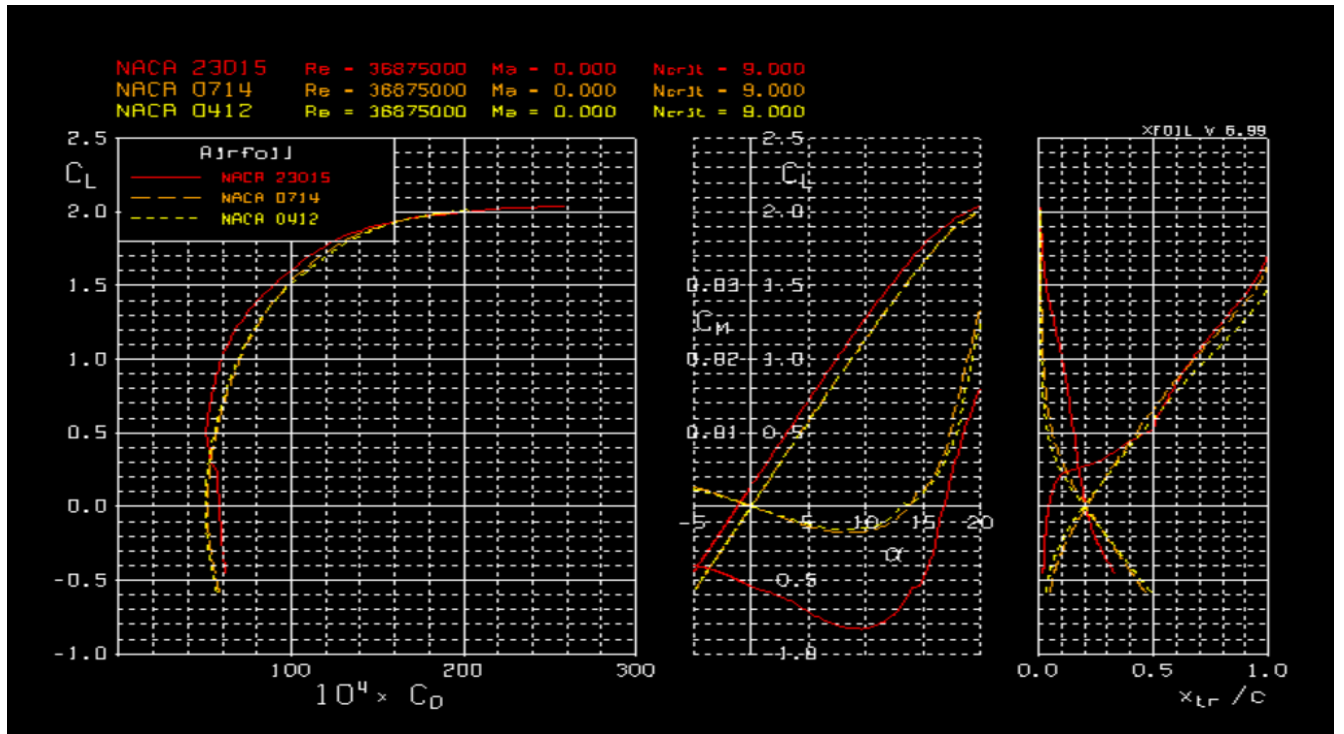


Figure 2: C_L vs. C_D , C_L vs. AoA for all three airfoils at $Re = 3.7e7$ (Cruise)

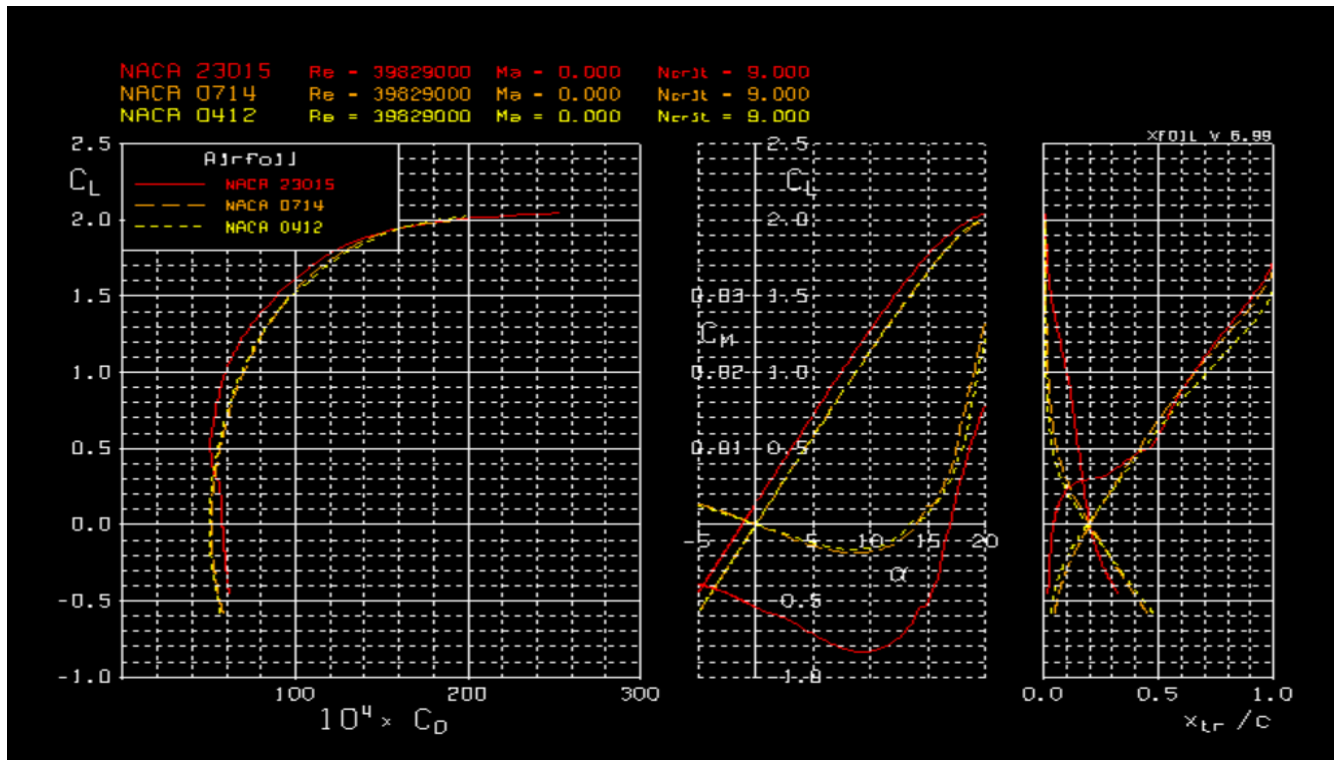


Figure 3: C_L vs. C_D , C_L vs. AoA for all three airfoils at $Re = 4.0e7$ (Descent)

Feature	NACA 0412	NACA 0714	NACA 23015
Camber	0% (symmetric)	Moderate (7%)	Forward, low camber
AoA_L=0	0°	-2.5°	-3.5°
Stall angle	High (stable)	Similar	Slightly earlier
Cm_c/4	0	-0.06 (est.)	-0.06 (est.)
Drag at cruise	Lowest	Moderate	Moderate-High

Figure 4: Table comparing the shapes, AoA_L=0, stall angle, Cm_c/4, and CD

Stall angle can be observed from the CL vs. AoA plots in figures 1-3. It differs based on Re.

As shown in all three figures, the NACA 23015 has a higher CL/CD ratio, meaning that it has greater efficiency at all three stages of flight. This also means that it has low-speed performance for short takeoffs and landings, ideal for rescue missions, and adverse weather conditions. The NACA 23015 also has a more negative Cm at lower AoA, which means it has a lower pitch moment, making it more stable. The boundary layer is also more favorable for the NACA 23015, meaning that the flow transitions later, allowing for smoother, and more predictable stall characteristics. For these reasons, our plane will utilize the NACA 23015 airfoil.

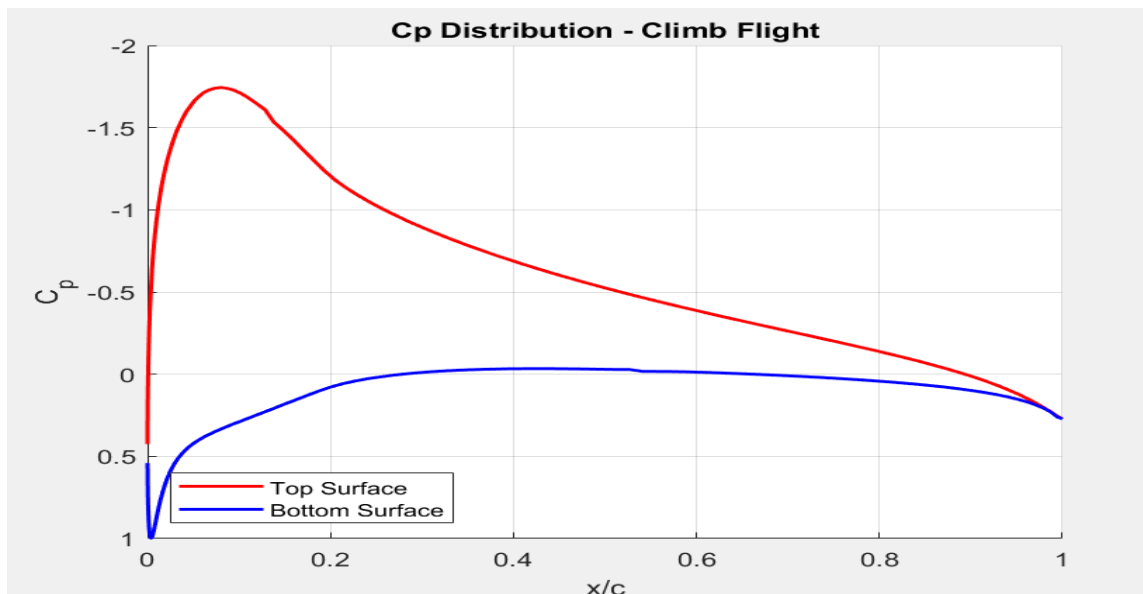


Figure 5: CP Distribution – Climb Flight

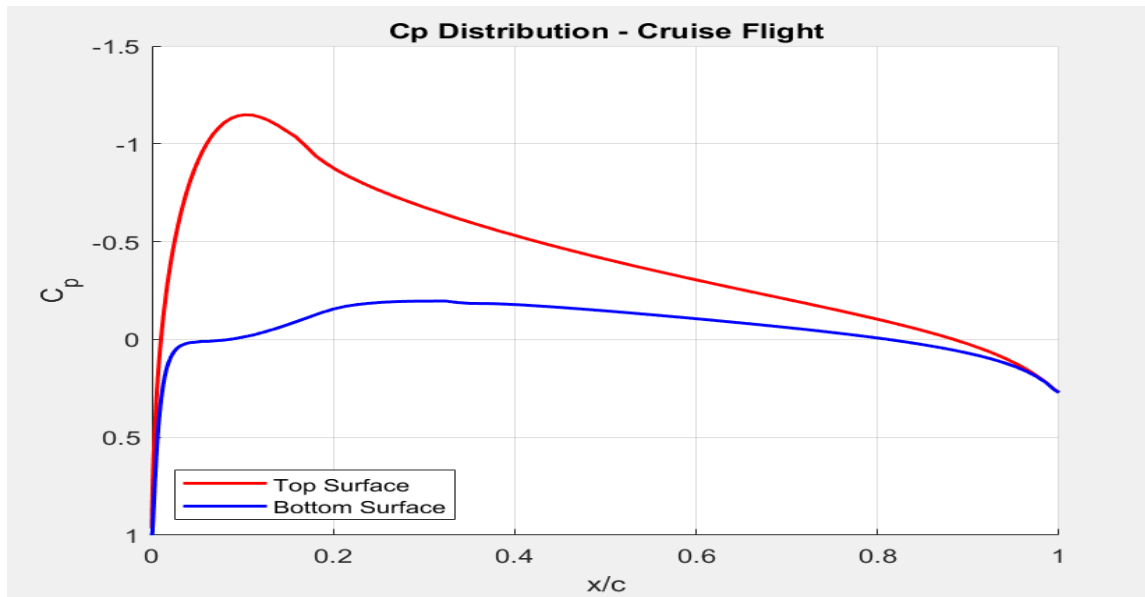


Figure 6: CP Distribution – Cruise Flight

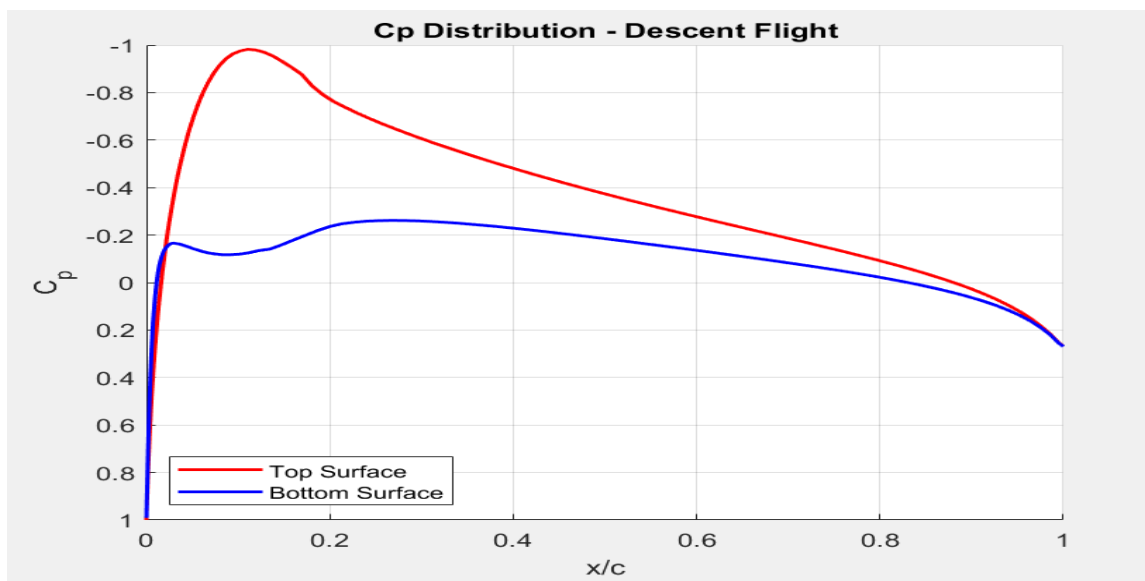


Figure 7: CP Distribution – Descent Flight

Part 2

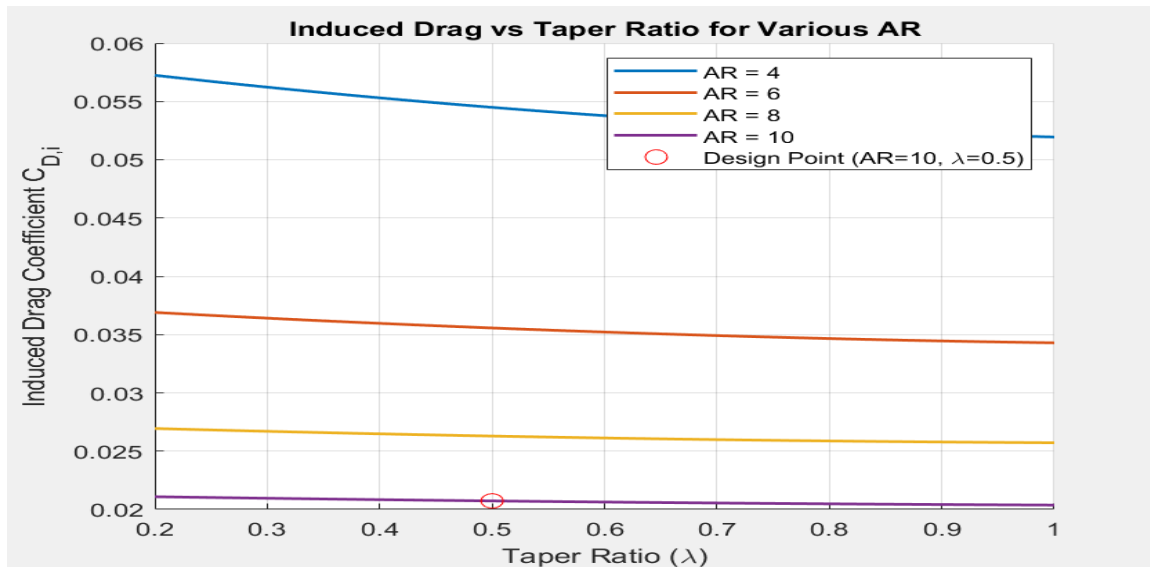


Figure 8: C_{Di} vs. Taper ratio for AR = 4, 6, 8, and 10.

As figure 7 shows, the wing used needs to have an AR of 10, and taper ratio of 0.5 for minimum induced drag.

Part 3

--- Cruise Performance Validation ---

Wing Area S: 241.07 m²

Root Chord: 6.43 m

Tip Chord: 3.21 m

Span: 50.00 m

Lift: 858256.36 N

Drag (Thrust Required): 24716.33 N

Pitching Moment Coefficient ($C_{m_{c/4}}$): -0.012

Torque at Wing Root Due to $C_{m_{c/4}}$: -138128.90 N*m

At 2 degrees AoA during cruise, the $C_{m_c/4}$ of this plane is -0.0119, and the torque on the wing at the cr is -138128.90 N*m. This is a milder pitching moment than normal standards, but fairly intense torque, which will require the wings to be reinforced heavily at the fuselage.

Part 4

In response to the mission statement, I selected the NACA 23015 airfoil for its favorable lift characteristics, moderate pitching moment, and high stability. The wing was optimized with an AR of 10 and a taper ratio of 0.5. Per figure 8, these characteristics allow for the lowest induced drag, giving the wing more efficient performance. The pitching moment coefficient $C_{m_c/4} = -0.0119$ generates a root torque of -138,128.90 N·m during cruise, which will require a reinforced wing-fuselage attachment at the quarter-chord location. This design ensures sufficient lift, while minimizing induced drag and enabling safe, reliable mission execution.

Note:

In section one of the project, I am comparing SC(2) – 0714, SC(2) – 0412, and NASA 23015, unfortunately, not of their files would properly run in Xfoil, so I had to choose different airfoils. In section two of this project, I am comparing NACA 0714, NACA 0412, and NACA 23015.